

single bundle, easing installation to the point of just hitting *Next* for each step and simply choosing the components.

Although this eases the task of installation considerably, manually installing and configuring the applications will help you understand the underlying system and configurations better.

This tutorial focuses on the manual installation of Apache and PHP. So, let's get started!

## Installing Apache

### Downloading the files

The Apache HTTP server website itself doesn't maintain an official binary build or an MSI installer for the newer versions and just hosts the source files. To get a binaries distribution, head over to <https://www.apachelounge.com/download/> and download the appropriate Zip file for your OS' architecture.

Also make sure the latest Microsoft Visual C++ runtime for your system is installed.

Extract the Zip file using your favourite file archiver utility like WinRAR, and store it in the desired location. For this tutorial, we are using `C:\Apache24` as the Apache home.

### Test run

By default, Apache is configured to listen to Port 80 of the server. You can test run the server by firing up your command prompt, then pressing the *Windows Key + R*, followed by typing 'cmd' in the *Run* box, and hitting *Enter*.

Then navigate to your installation directory and enter the bin folder that houses the executable for the server; for me it is at `C:\Apache24\bin`.

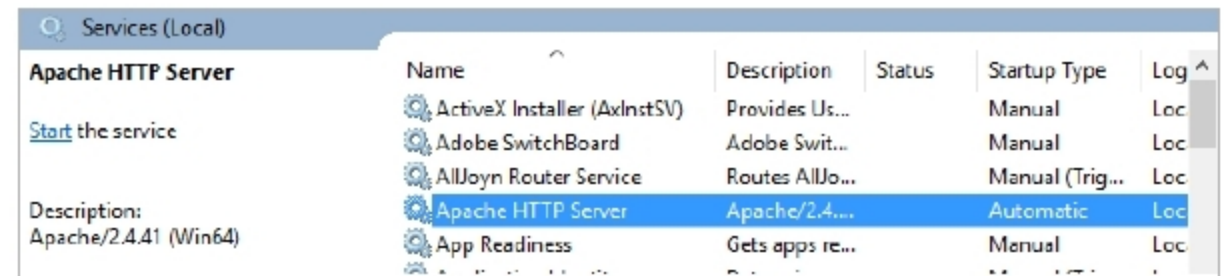
```
cd C:\Apache24\bin
```

Once you have entered the bin folder, run the following command:

```
httpd.exe
```

```
C:\Apache24\bin>httpd.exe
AH00558: httpd.exe: Could not reliably determine the server's fully qualified domain name, using fe80::8d36:9aed:e690:f9a5. Set the 'ServerName' directive globally to suppress this message
```

Figure 1: Apache warns about server's FQDN



| Name                         | Description     | Status | Startup Type    | Log |
|------------------------------|-----------------|--------|-----------------|-----|
| ActiveX Installer (AxInstSV) | Provides Us...  |        | Manual          | Loc |
| Adobe SwitchBoard            | Adobe Swit...   |        | Manual          | Loc |
| AllJoyn Router Service       | Routes AllJo... |        | Manual (Trig... | Loc |
| Apache HTTP Server           | Apache/2.4...   |        | Automatic       | Loc |
| App Readiness                | Gets apps re... |        | Manual          | Loc |

Figure 2: Apache listed as a service

This will trigger the Windows security to ask for your permission. You can choose to allow Apache to be accessed over a private network like your home network or public networks. It's up to you to make that choice. If this is a personal development system, I recommend that you just select *Private networks* and click on *Allow access*.

Once you do that, you can see a similar output on the command prompt (Figure 1).

This means that Apache has started running correctly, but a domain name was not set for Apache, which is fine for now, as we are just testing it and not deploying it on a real server for the world to access.

To get rid of this message, go to the *conf* directory in the *Apache24* installation folder. Open the file *httpd.conf* with your favourite text editor and find the line that says *# ServerName www.example.com:80*. Remove the hash # and replace *www.example.com* with *localhost*.

After this is done, it should look like what follows:

```
#
# If your host doesn't have a
# registered DNS name, enter its IP
# address here.
#
ServerName localhost:80
```

Now open up a browser and the URL: `http://localhost`. This should open up a page saying "This works!" which

means just that. Apache was successfully able to be run on your computer.

### Running as a server

You can't expect to fire up a command prompt every time you'd like to play around with your Web server. In order to automatically start it when you start your computer and run it in the background, you need to register it as a service.

Exit the application first by pressing *Ctrl+C* in the command prompt and close the command prompt.

Since we are going to install the application, we need administrative privileges. So, run the command prompt as an administrator first and navigate to the Apache bin directory once again. Then run the following command:

```
httpd.exe -k install -n "Apache HTTP Server"
```

This will install the Apache service. Run the following to check if it has been installed correctly:

```
services.msc
```

You should be able to find Apache HTTP Server in the list of services active on your computer.

The startup type for Apache should be set as *Automatic*.

Apache is now active on your computer. Let us configure Apache to work according to our needs.